

Lab: Active Inference Control vs. Classical Motor Control

Objective

Compare PID control, Model Predictive Control (MPC), and Active Inference control for a robotic manipulator task. You will design controllers for a 2-DOF planar arm performing a reaching task, analyze their behavior under perturbation, and evaluate how Active Inference unifies perception and action in a single objective.

Prerequisites

- Understanding of basic control theory (PID, feedback loops)
- Familiarity with robot kinematics (forward/inverse kinematics)
- Module 04: Cognition (world models that support action selection)

Part 1: Task Specification

A 2-DOF planar robot arm must reach from an initial configuration ($q_1=0$, $q_2=0$) to a target end-effector position ($x_{\text{target}}=0.5\text{m}$, $y_{\text{target}}=0.3\text{m}$). The arm has link lengths $L_1=0.4\text{m}$ and $L_2=0.3\text{m}$.

1. Write the forward kinematics equations mapping joint angles (q_1 , q_2) to end-effector position (x , y).
2. Compute the target joint angles using inverse kinematics.
3. Define the observation model: the robot observes joint angles via encoders (with noise $\sigma_q = 0.01$ rad) and end-effector position via a camera (with noise $\sigma_{xy} = 0.005$ m).

Write your response here

Part 2: PID Controller Design

Design a joint-space PID controller for the reaching task:

1. Specify the error signal: $e(t) = q_{\text{target}} - q_{\text{measured}}$.
2. Choose PID gains (K_p , K_i , K_d) and justify your choices.
3. Describe the controller output: torque commands $\tau = K_p e + K_i \text{integral}(e) + K_d \frac{de}{dt}$.
4. What happens if an unexpected load is placed on the end-effector during the reach?

How does the PID controller respond?

Write your response here

Part 3: Active Inference Controller Design

Design an Active Inference controller for the same reaching task:

1. **Generative model:** Specify prior preferences over observations (desired joint angles and end-effector position).
2. **Prediction errors:** Define sensory prediction errors (difference between predicted and observed joint angles) and active prediction errors (difference between desired and predicted proprioceptive states).
3. **Action as reflex arc:** In Active Inference, motor commands arise from proprioceptive prediction errors -- the arm moves to fulfill its own predictions. Write the update equation:
4. $u = -dF/da$ (action minimizes free energy by changing sensory input)
5. How does the Active Inference controller handle the unexpected load? Compare with PID.

Write your response here

Part 4: Perturbation Analysis

Compare all three controllers (PID, MPC, Active Inference) under three perturbation scenarios:

Scenario	PID Response	MPC Response	Active Inference Response
External force on end-effector			
Encoder noise increases 10x			
Joint 2 motor loses 50% torque			

For each scenario, describe qualitatively how the controller adapts (or fails to adapt).

Write your response here

Part 5: Unified Perception-Action Pseudocode

Write pseudocode for a complete Active Inference control loop that unifies perception and action:

```
function active_inference_control(dt):  
    while not reached_target:  
        # Perception: update beliefs about joint angles  
        mu_q += dt * (d_mu_q - kappa * prediction_error_proprio)  
  
        # Perception: update beliefs about end-effector  
        mu_x += dt * (d_mu_x - kappa * prediction_error_visual)  
  
        # Action: generate motor commands from prediction errors  
        torque = -kappa_a * (mu_q - q_desired)  
  
        # Apply torque and read new sensors  
        apply(torque)  
        q_obs, x_obs = read_sensors()
```

Extend this pseudocode to handle the perturbation scenarios from Part 4.

Write your response here

Summary Table

Feature	PID	MPC	Active Inference
Objective function	Error minimization	Cost over horizon	Free energy minimization
Perception integration	None (assumes known state)	State estimator separate	Unified with control
Adaptation	Requires gain scheduling	Requires model updates	Precision learning
Computational cost	_____	_____	_____
Robustness to noise	_____	_____	_____

References

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- Lanillos, P., et al. (2021). Active Inference in Robotics and Artificial Agents. *Frontiers in Neurorobotics*.
- Oliver, G., Lanillos, P., & Cheng, G. (2021). An empirical study of active inference on a humanoid robot. *IEEE Transactions on Cognitive and Developmental Systems*.